

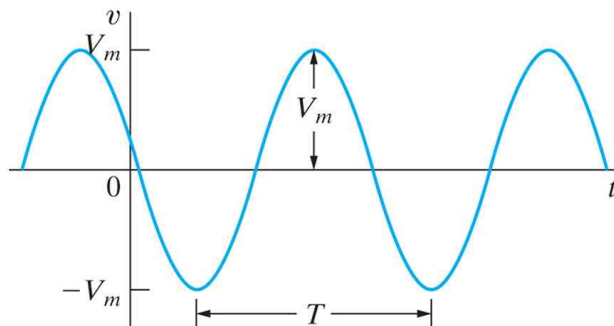
Chap.9 Sinusoidal Steady-State Analysis

§9.0 Why sinusoidal waves are widely used in engineering?

- (i) easy to make
- (ii) many sinusoidal phenomena in the real world
- (iii) easy to consider mathematically
(differentiation of a sinusoidal wave is also a sinusoidal wave,
integration of a sinusoidal wave is also a sinusoidal wave)
- (iv) Fourier Analysis
(a signal can be represented as summation of sinusoidal waves)

§9.1 The Sinusoidal Source

- A sinusoidal voltage



$$v(t) = V_m \cos(\omega t + \phi) = V_m \cos(2\pi f t + \phi)$$

amplitude : V_m [V]

period : T [sec]

frequency : f [Hz] or [1/sec]

angular frequency : $\omega = 2\pi f = \frac{2\pi}{T}$ [rad/sec]

phase angle : ϕ [rad]

- the general equation for a sinusoidal source

voltage source : $v(t) = V_m \cos(\omega t + \phi) = V_m \cos(2\pi f t + \phi)$

current source : $i(t) = I_m \cos(\omega t + \phi) = I_m \cos(2\pi f t + \phi)$

- the rms value of a sinusoidal voltage source

※ rms: square root of the mean value of the squared function
or root-mean-square value

$$\begin{aligned} V_{\text{rms}} &= \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} v^2(t) dt} \\ &= \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} V_m^2 \cos^2(\omega t + \phi) dt} \\ &= \frac{V_m}{\sqrt{2}} \end{aligned}$$

- ※ The value of AC (alternating current) varies with time.

So we can not represent the voltage of AC as DC (Direct current) voltage.

The effective value (i.e., rms value) of any periodic current is equal to the value of direct current which, flowing through an R ohm resistor, delivers the same average power to the resistor as does the periodic current.

(Example) $v(t) = 300 \cos(120 \pi t + 30^\circ)$ V

- a) period

$$\omega = 120 \pi \text{ [rad/sec]}$$

$$\therefore T = \frac{2\pi}{\omega} = \frac{1}{60} = 16.667 \text{ [msec]}$$

- b) frequency

$$f = \frac{1}{T} = 60 \text{ [Hz]}$$

- c) $v(t)$ at $t = 2.778 \text{ msec}$

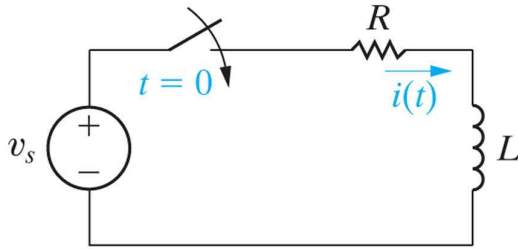
$$\begin{aligned} 120 \pi t &= 120 \pi \times 2.778 \times 10^{-3} = 1.047 \text{ rad} \\ &= 1.047 \times 180 / \pi = 60^\circ \end{aligned}$$

$$\therefore v(2.778 \text{ msec}) = 300 \cos(60^\circ + 30^\circ) = 0 \text{ V}$$

- d) root mean square value

$$V_{\text{rms}} = V_m / \sqrt{2} = 300 / \sqrt{2} = 212.13 \text{ V}$$

§9.2 The Sinusoidal Response



$$v_s(t) = V_m \cos(\omega t + \phi)$$

$$L \frac{di(t)}{dt} + Ri(t) = V_m \cos(\omega t + \phi) \quad \dots\dots\dots \text{Eq.(\#)}$$

$$i(t) = \frac{-V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\phi - \theta) e^{-(R/L)t} \quad [\text{transient response}]$$

$$+ \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\omega t + \phi - \theta) \quad [\text{steady-state response}]$$

$$\text{where } \theta = \tan^{-1}(\omega L/R)$$

※ It is not easy to solve the second-order nonhomogeneous differential equation (#). [Engineering Mathematics]

● WE FOCUS ON THE STEADY-STATE PORTION IN THIS CHAPTER.

1. The steady-state response is a sinusoidal function.
2. The frequency, ω , of a sinusoidal response is the same as the frequency of the sinusoidal source driving the circuit.
3. The amplitude and phase angle of the response are usually different from those of the source.

※ The Response of a Linear Systems

§9.3 The Phasor

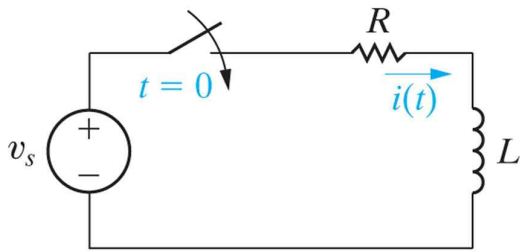
● Euler's Identity

$$e^{\pm j\theta} = \cos \theta \pm j \sin \theta$$

$$\cos \theta = \text{Re}\{e^{j\theta}\} \text{ and } \sin \theta = \text{Im}\{e^{j\theta}\}$$

(※ $e^{j\pi} + 1 = 0$: Some mathematicians say this equation is the most beautiful eq. in the world, since it includes e , π , 0, 1 and j .)

● Input and output of a Linear System



Handwritten notes illustrating the input and output of a linear system for an RL circuit.

Case 1: Cosine Input

Input voltage: $v(t) = V_m \cos(\omega t + \phi)$

Circuit diagram shows a voltage source $v(t)$ in series with a resistor R and an inductor L .

Output current: $i(t) = I_m \cos(\omega t + \theta)$

Case 2: Sine Input

Input voltage: $v(t) = V_m \sin(\omega t + \phi)$

Circuit diagram shows a voltage source $v(t)$ in series with a resistor R and an inductor L .

Output current: $i(t) = I_m \sin(\omega t + \theta)$

Superposition (Linear System)

Input voltage: $v(t) = V_m \cos(\omega t + \phi) + j V_m \sin(\omega t + \phi)$

Input voltage in complex exponential form: $= V_m e^{j(\omega t + \phi)}$

Circuit diagram shows two voltage sources in series: $V_m \cos(\omega t + \phi)$ and $j V_m \sin(\omega t + \phi)$, in series with a resistor R and an inductor L .

Output current in complex exponential form: $i(t) = I_m \cos(\omega t + \theta) + j I_m \sin(\omega t + \theta) = I_m e^{j(\omega t + \theta)}$

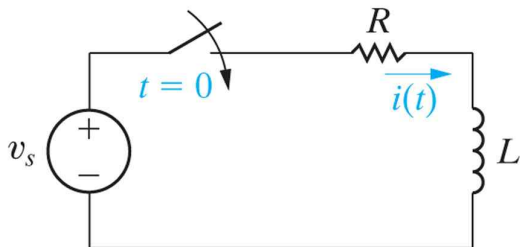
Real part of the output: $\text{Re}\{I_m e^{j(\omega t + \theta)}\}$

Real part of the input: $\text{Re}\{V_m e^{j(\omega t + \phi)}\}$

- Summary

input of a linear system	output of the system
$v(t) = V_m \cos(\omega t + \phi)$	$i(t) = I_m \cos(\omega t + \theta)$
$v(t) = V_m \sin(\omega t + \phi)$	$i(t) = I_m \sin(\omega t + \theta)$
$v(t) = V_m \cos(\omega t + \phi)$ $+ j V_m \sin(\omega t + \phi)$ $= V_m e^{j(\omega t + \phi)}$	$i(t) = I_m \cos(\omega t + \theta)$ $+ j I_m \sin(\omega t + \theta)$ $= I_m e^{j(\omega t + \theta)}$
$v(t) = V_m \cos(\omega t + \phi)$ $= \operatorname{Re} \{ V_m e^{j(\omega t + \phi)} \}$	$i(t) = I_m \cos(\omega t + \theta)$ $= \operatorname{Re} \{ I_m e^{j(\omega t + \theta)} \}$

● RL circuit with trigonometric function



$$v_s(t) = V_m \cos(\omega t + \phi) \quad \text{Let } i(t) = I_m \cos(\omega t + \theta)$$

$$L \frac{di(t)}{dt} + Ri(t) = V_m \cos(\omega t + \phi)$$

$$L \frac{d}{dt} \{ I_m \cos(\omega t + \theta) \} + RI_m \cos(\omega t + \theta) = V_m \cos(\omega t + \phi)$$

$$-\omega L I_m \sin(\omega t + \theta) + RI_m \cos(\omega t + \theta) = V_m \cos(\omega t + \phi)$$

Now, how to solve ? Easy to solve ?

(※ It takes long time to solve the equation, and we must use many trigonometric identities.)

※ Some Useful Trigonometric Identities

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\cos(\alpha \pm 90^\circ) = \mp \sin \alpha$$

$$\sin(\alpha \pm 90^\circ) = \pm \cos \alpha$$

$$\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha + \beta) + \frac{1}{2} \cos(\alpha - \beta)$$

$$\sin \alpha \sin \beta = \frac{1}{2} \cos(\alpha - \beta) - \frac{1}{2} \cos(\alpha + \beta)$$

$$\sin \alpha \cos \beta = \frac{1}{2} \sin(\alpha + \beta) + \frac{1}{2} \sin(\alpha - \beta)$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\sin^2 \alpha = \frac{1}{2}(1 - \cos 2\alpha)$$

$$\cos^2 \alpha = \frac{1}{2}(1 + \cos 2\alpha)$$

$$\sin \alpha = \frac{e^{j\alpha} - e^{-j\alpha}}{j2}$$

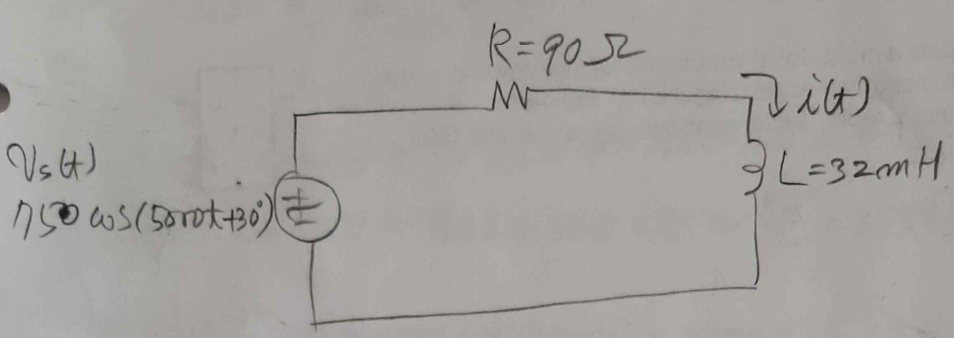
$$\cos \alpha = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$$

$$e^{\pm j\alpha} = \cos \alpha \pm j \sin \alpha$$

$$A \cos \alpha + B \sin \alpha = \sqrt{A^2 + B^2} \cos \left(\alpha + \tan^{-1} \frac{-B}{A} \right)$$

(Example) Find the output signal $i(t)$ using the trigonometric method.

①



$V_s(t) = 1750 \cos(5000t + 30^\circ)$
 $R = 90 \Omega$
 $L = 32 \text{ mH}$
 $i(t)$

$$V_s(t) = 1750 \cos(5000t + 30^\circ)$$

$$i(t) = I_m \cos(5000t + \theta)$$

$$L \frac{di}{dt} + Ri = V_s(t)$$

$$0.032 \frac{d}{dt} [I_m \cos(5000t + \theta)] + 90 I_m \cos(5000t + \theta) = 1750 \cos(5000t + 30^\circ)$$

$$= 1750 \cos(5000t + 30^\circ)$$

$$-5000 \times 0.032 I_m \sin(5000t + \theta) + 90 I_m \cos(5000t + \theta) = 1750 \cos(5000t + 30^\circ)$$

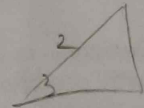
$$= 1750 \cos(5000t + 30^\circ)$$

$$-160 I_m [\sin 5000t \cos \theta + \cos 5000t \sin \theta] + 90 I_m [\cos 5000t \cos \theta - \sin 5000t \sin \theta] - 1750 [\cos 5000t \cos 30^\circ - \sin 5000t \sin 30^\circ] = 0$$

$$[-160 I_m \cos \theta - 90 I_m \sin \theta + 1750 \sin 30^\circ] \sin 5000t + [-160 I_m \sin \theta + 90 I_m \cos \theta - 1750 \cos 30^\circ] \cos 5000t = 0$$

$$\sin 30^\circ = 0.5 \quad \cos 30^\circ = \frac{\sqrt{3}}{2}$$

②



$$\therefore +160 I_m \cos \theta + 90 I_m \sin \theta = 750 \times 0.5 = 375 \quad \dots (*)$$

$$-160 I_m \sin \theta + 90 I_m \cos \theta = 750 \times \frac{\sqrt{3}}{2} = 375\sqrt{3} \quad \dots (**)$$

$$\begin{aligned} &\rightarrow 25600 I_m^2 \cos^2 \theta + 8100 I_m^2 \sin^2 \theta + 28800 I_m^2 \cos \theta \sin \theta \\ &= 140625 \end{aligned}$$

$$\begin{aligned} +) & 25600 I_m^2 \sin^2 \theta + 8100 I_m^2 \cos^2 \theta - 28800 I_m^2 \sin \theta \cos \theta \\ &= 421875 \end{aligned}$$

$$\begin{aligned} &\frac{25600 I_m^2 (\cos^2 \theta + \sin^2 \theta)}{=1} + \frac{8100 I_m^2 (\sin^2 \theta + \cos^2 \theta)}{=1} \\ &= 562500 \end{aligned}$$

$$\therefore 25600 I_m^2 + 8100 I_m^2 = 562500$$

$$33700 I_m^2 = 562500$$

$$\therefore I_m^2 = 16.69139$$

$$\therefore I_m = 4.0855$$

(3)

$$(*) \quad 160 I_m \cos \theta + 90 I_m \sin \theta = 315 \quad \text{--- (a)}$$

$$(**) \quad -160 I_m \sin \theta + 90 I_m \cos \theta = 315\sqrt{3}$$



$$90 I_m \cos \theta - 160 I_m \sin \theta = 315\sqrt{3} \quad \text{--- (b)}$$

$$(a) \times 9$$

$$1440 I_m \cos \theta + 810 I_m \sin \theta = 3375$$

$$(b) \times 16$$

$$1440 I_m \cos \theta - 2560 I_m \sin \theta = 6500\sqrt{3} \quad \text{---}$$

$$3370 I_m \sin \theta = -1017.305$$

$$\therefore \sin \theta = \frac{-1017.305}{3370 \times 4.0855 I_m}$$

$$\therefore \theta = -0.509677$$

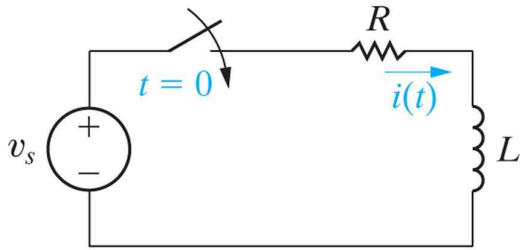
$$\therefore \theta = \sin^{-1}(-0.509677)$$

$$= -30.6423^\circ$$

$$\therefore i(t) = I_m \cos(5000t + \theta)$$

$$= 4.0855 \cos(5000t - 30.6423^\circ)$$

● RL circuit with Euler's Identity



$$v_s(t) = V_m e^{j(\omega t + \phi)}, \text{ Let } i(t) = I_m e^{j(\omega t + \theta)}$$

$$L \frac{di(t)}{dt} + R i(t) = v_s(t)$$

$$L \frac{d}{dt} I_m e^{j(\omega t + \theta)} + R I_m e^{j(\omega t + \theta)} = V_m e^{j(\omega t + \phi)}$$

$$j\omega L I_m e^{j(\omega t + \theta)} + R I_m e^{j(\omega t + \theta)} = V_m e^{j(\omega t + \phi)}$$

$$[j\omega L + R] I_m e^{j\theta} e^{j\omega t} = V_m e^{j\phi} e^{j\omega t}$$

$$[j\omega L + R] I_m e^{j\theta} = V_m e^{j\phi} \quad (*)$$

$$I_m e^{j\theta} = \frac{V_m}{R + j\omega L} e^{j\phi}$$

$$I_m e^{j\theta} = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} e^{j[\phi - \tan^{-1}(\omega L/R)]}$$

$$\therefore I_m = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}}, \quad \theta = \phi - \tan^{-1}(\omega L/R)$$

$$\therefore i(t) = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos \left[\omega t + \phi - \tan^{-1} \left(\frac{\omega L}{R} \right) \right]$$

※ After eliminating $e^{j\omega t}$ from both sides, we can get Eq.(*).

※ Eq.(*) : NO time (t) terms in the equation.

This equation includes not time (t) but frequency (ω).

From now our playground is the frequency domain,

not the time domain. (a very important concept).

From now we will analyze a circuit in the frequency domain.

● Phasor transform

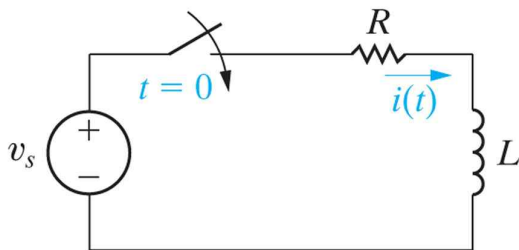
time domain representation

$$\begin{aligned}
 v(t) &= V_m \cos(\omega t + \phi) \\
 &= V_m \operatorname{Re}\{e^{(j\omega t + \phi)}\} \\
 &= V_m \operatorname{Re}\{e^{j\omega t} e^{j\phi}\} \\
 &= \operatorname{Re}\{V_m e^{j\phi} e^{j\omega t}\}
 \end{aligned}$$

frequency domain representation

$$\begin{aligned}
 \mathbf{V} &= P\{V_m \cos(\omega t + \phi)\} \quad (\text{* phasor transform}) \\
 &= V_m e^{j\phi} = V_m \angle \phi \quad (\text{PHASOR [important concept]}) \\
 &= V_m (\cos \phi + j \sin \phi) \quad (\text{* Phasors are COMPLEX numbers.})
 \end{aligned}$$

● RL circuit with the phasor



Reuse Eq.(*)

$$[R + j\omega L] I_m e^{j\theta} = V_m e^{j\phi}$$

$$\text{let } \mathbf{I} = I_m e^{j\theta} = I_m \angle \theta \quad (\text{phasor})$$

$$\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi \quad (\text{phasor})$$

$$[R + j\omega L] I_m \angle \theta = V_m \angle \phi \quad \text{or} \quad [R + j\omega L] \mathbf{I} = \mathbf{V}$$

$$I_m \angle \theta = \frac{V_m \angle \phi}{R + j\omega L} \quad \text{or} \quad \mathbf{I} = \frac{\mathbf{V}}{R + j\omega L}$$

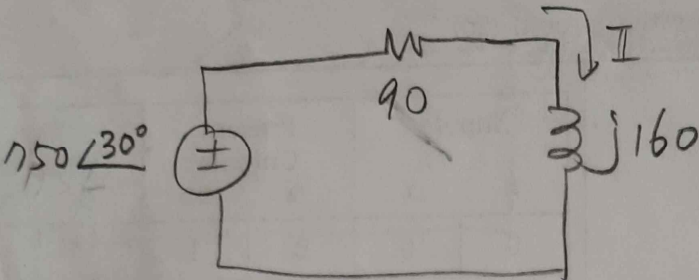
$$\text{let } \mathbf{Z} = R + j\omega L \quad (\text{Impedance})$$

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}} \quad (\text{Ohm's Law})$$

(Example) Phasor method

④

(b)

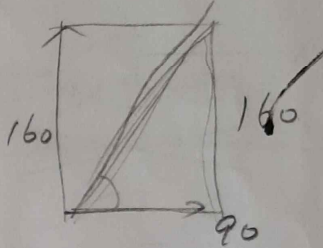


$$I = \frac{750 \angle 30^\circ}{90 + j160}$$

$$= \frac{750 \angle 30^\circ}{\sqrt{8100 + 25600} \angle 60.6422^\circ}$$

$$= \frac{750 \angle 30^\circ}{183.51156 \angle 60.6422^\circ}$$

$$= 4.0855 \angle -30.6422^\circ$$

$$\therefore i(t) = 4.0855 \cos(5000t - 30.6422^\circ)$$


● time domain and frequency domain

time domain	frequency domain (phasor)
$v_s(t) = V_m \cos(\omega t + \phi)$	$\mathbf{V} = V_m e^{j\phi} = I_m \angle \phi$
$i(t) = I_m \cos(\omega t + \theta)$	$\mathbf{I} = I_m e^{j\theta} = I_m \angle \theta$
$R, L, d/dt$	$R, L, j\omega$
$L \frac{di(t)}{dt} + Ri(t) = v_s(t)$	$(R + j\omega L) I_m \angle \theta = V_m \angle \phi$
differential equation	algebraic equation, but complex numbers
.....	$I_m \angle \theta = \frac{V_m \angle \phi}{R + j\omega L}$
$i(t) = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos[\omega t + \phi - \tan^{-1}(\frac{\omega L}{R})]$	

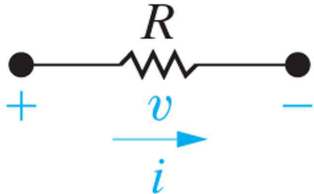
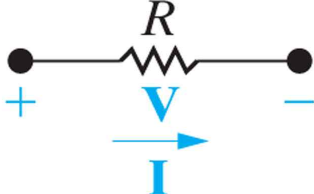
(Example) $y_1(t) = 20 \cos(\omega t - 30^\circ)$, $y_2(t) = 40 \cos(\omega t + 60^\circ)$

$y = y_1 + y_2$ (trigonometric method)	$y = y_1 + y_2$ (phasor method)
$y_1(t) = 20 \cos(\omega t - 30^\circ)$ $y_2(t) = 40 \cos(\omega t + 60^\circ)$ using trigonometric theorems $y_1 = 20 \cos \omega t \cos 30^\circ + 20 \sin \omega t \sin 30^\circ$ $y_2 = 40 \cos \omega t \cos 60^\circ - 40 \sin \omega t \sin 60^\circ$ $y = (20 \cos 30^\circ + 40 \cos 60^\circ) \cos \omega t$ $+ (20 \sin 30^\circ - 40 \sin 60^\circ) \sin \omega t$ $= 37.32 \cos \omega t - 24.64 \sin \omega t$ $= 44.72 \left(\frac{37.32}{44.72} \cos \omega t - \frac{24.64}{44.72} \sin \omega t \right)$ $= 44.72 (\cos 33.43^\circ \cos \omega t - \sin 33.43^\circ \sin \omega t)$ $= 44.72 \cos(\omega t + 33.43^\circ)$	$y_1(t) = 20 \cos(\omega t - 30^\circ)$ $y_2(t) = 40 \cos(\omega t + 60^\circ)$ using phasors $\mathbf{Y}_1 = 20 \angle -30^\circ$ $\mathbf{Y}_2 = 40 \angle 60^\circ$ $\mathbf{Y} = \mathbf{Y}_1 + \mathbf{Y}_2$ $= 20 \angle -30^\circ + 40 \angle 60^\circ$ $= (17.32 - j10) + (20 + j34.64)$ $= 37.32 + j24.64$ $= 44.72 \angle 33.43^\circ$ $\therefore y = 44.72 \cos(\omega t + 33.43^\circ)$

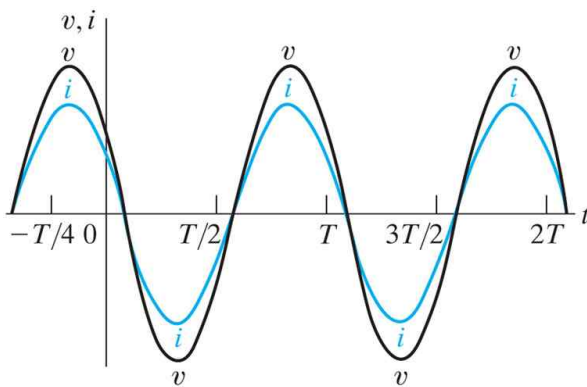
※ The superiority of the phasor approach for adding sinusoidal functions should be apparent. It requires the ability to move back and forth between the polar and rectangular forms of complex numbers.

§9.4 Passive Circuit Elements in the Frequency Domain

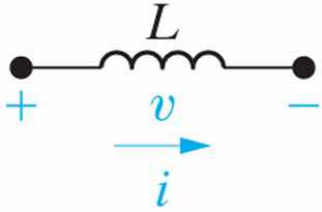
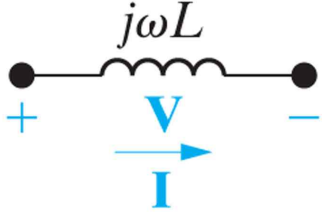
● The V-I Relationship for a Resistor

time domain	frequency domain [phasor]
	
$i(t) = I_m \cos(\omega t + \theta_i)$ $v(t) = R I_m \cos(\omega t + \theta_i)$ $\therefore v(t) = R i(t)$	$I = I_m \angle \theta_i$ $V = R I_m \angle \theta_i$ $\therefore V = R I$

※ Resistor: The voltage and current are in phase [※ same phase].

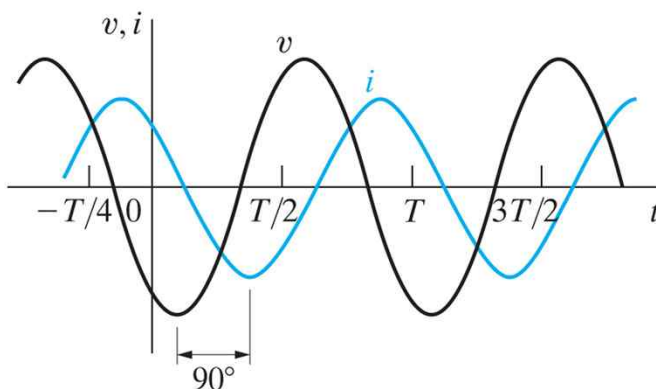


● The V-I Relationship for an Inductor

time domain	frequency domain [phasor]
	
$i(t) = I_m \cos(\omega t + \theta_i)$ $v(t) = L \frac{di(t)}{dt}$ $= -\omega L I_m \sin(\omega t + \theta_i)$ $= -\omega L I_m \cos(\omega t + \theta_i - 90^\circ)$ $\therefore$ impossible to represent like the Ohm's law (differentiation is needed) (not simple)	$I = I_m \angle \theta_i$ $V = -\omega L I_m \angle (\theta_i - 90^\circ)$ $= -\omega L I_m e^{j(\theta_i - 90^\circ)}$ $= -\omega L I_m e^{-j90^\circ} e^{j\theta_i}$ $= j\omega L I_m e^{j\theta_i}$ $= j\omega L (I_m \angle \theta_i)$ $\therefore V = (j\omega L) I$ (a kind of the Ohm's law) (simple)

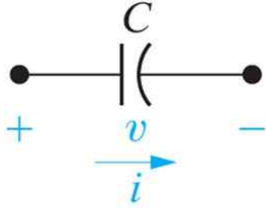
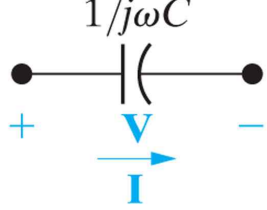
$$V = (j\omega L) (I_m \angle \theta_i) = (\omega L \angle 90^\circ) (I_m \angle \theta_i)$$

$$= \omega L I_m \angle (\theta_i + 90^\circ)$$



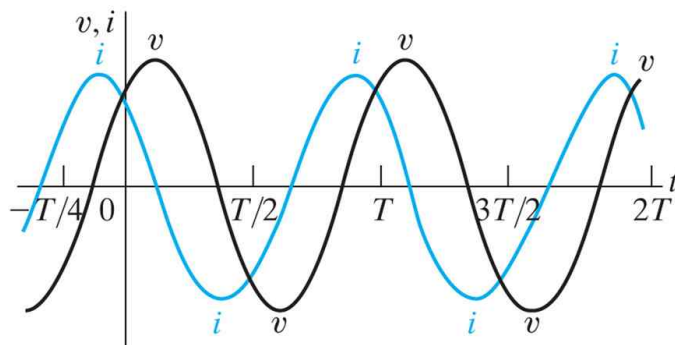
- ※ voltage leading current or current lagging voltage by 90°
- ※ Phase difference in frequency domain means
time difference in time domain.

● The V-I Relationship for a Capacitor

time domain	frequency domain [phasor]
	
$v(t) = V_m \cos(\omega t + \theta_v)$ $i(t) = C \frac{dv(t)}{dt}$ $= -\omega C V_m \sin(\omega t + \theta_v)$ $= -\omega C V_m \cos(\omega t + \theta_v - 90^\circ)$ $\therefore$ impossible to represent like the Ohm's law (differentiation is needed) (not simple)	$V = V_m \angle \theta_v$ $I = -\omega C V_m \angle (\theta_v - 90^\circ)$ $= -\omega C V_m e^{j(\theta_v - 90^\circ)}$ $= -\omega C V_m e^{-j90^\circ} e^{j\theta_v}$ $= j\omega C V_m e^{j\theta_v}$ $= j\omega C (V_m \angle \theta_v)$ $\therefore I = (j\omega C) V$ $\therefore V = \frac{1}{j\omega C} I$ (a kind of the Ohm's law) (simple)

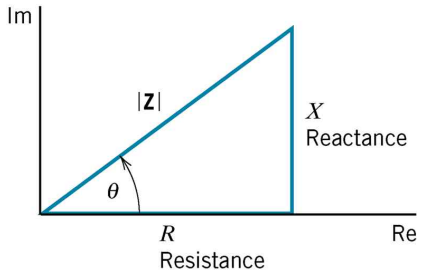
$$V = \frac{1}{j\omega C} (I_m \angle \theta_i) = \left(\frac{1}{\omega C} \angle -90^\circ \right) (I_m \angle \theta_i)$$

$$= \frac{I_m}{\omega C} \angle (\theta_i - 90^\circ)$$



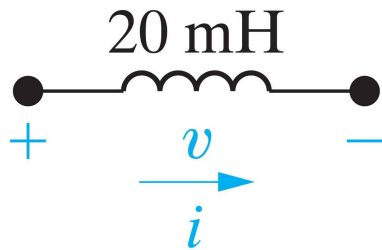
- ※ current leading voltage or voltage lagging current by 90°
- ※ peak time of voltage and current is different in RL and RC.
[In resistor circuits, voltage and current are in phase.]

● Impedance and Reactance

resistor $V = RI$	Impedance Z (unit: Ω) $V = ZI$ 
inductor $V = (j\omega L)I$	
capacitor $V = \frac{1}{j\omega C}I$ $= -j\frac{1}{\omega C}I$	

※ Reactance : ωL , $-\frac{1}{\omega C}$

(Example) $i(t) = 10 \cos(10,000t + 30^\circ)$ mA



(i) impedance

$$j\omega L = j(10,000)(0.02) = j200 [\Omega]$$

(ii) the phasor voltage

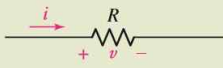
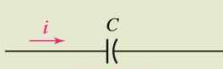
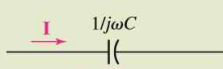
$$\begin{aligned} V &= (j\omega L)I = j200 \times 0.01 \angle 30^\circ \\ &= (200 \angle 90^\circ) \times (0.01 \angle 30^\circ) \\ &= 2 \angle 120^\circ [\text{V}] \end{aligned}$$

(iii) the steady-state expression for $v(t)$

$$v(t) = 2 \cos(10,000t + 120^\circ) [\text{V}]$$

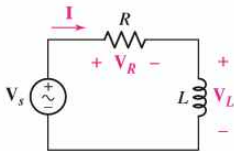
※ If we want to calculate this $v(t)$ in the time domain, it takes long time.

(Summary) Comparison of Time-Domain and Frequency-Domain
Voltage-Current Expressions

Time Domain		Frequency Domain	
	$v = Ri$	$\mathbf{V} = R\mathbf{I}$	
	$v = L \frac{di}{dt}$	$\mathbf{V} = j\omega L\mathbf{I}$	
	$v = \frac{1}{C} \int i dt$	$\mathbf{V} = \frac{1}{j\omega C}\mathbf{I}$	

§9.5 Kirchhoff's Laws in the Frequency Domain

	time domain	frequency domain
KVL	$\sum_{n=1}^N v_n(t) = 0$	$\sum_{n=1}^N \mathbf{V}_n = 0$
KCL	$\sum_{n=1}^N i_n(t) = 0$	$\sum_{n=1}^N \mathbf{I}_n = 0$
	Real number equation, complex calculation	complex number equation, easy calculation



■ FIGURE 10.16 The series RL circuit with a phasor voltage applied.

Now let us look briefly at the series RL circuit that we have considered several times before. The circuit is shown in Fig. 10.16, and a phasor current and several phasor voltages are indicated. We may obtain the desired response, a time-domain current, by first finding the phasor current. From Kirchhoff's voltage law,

$$\mathbf{V}_R + \mathbf{V}_L = \mathbf{V}_s$$

and using the recently obtained V-I relationships for the elements, we have

$$R\mathbf{I} + j\omega L\mathbf{I} = \mathbf{V}_s$$

The phasor current is then found in terms of the source voltage $\mathbf{V}_s$:

$$\mathbf{I} = \frac{\mathbf{V}_s}{R + j\omega L}$$

Let us select a source-voltage amplitude of V_m and phase angle of 0° . Thus,

$$\mathbf{I} = \frac{V_m \angle 0^\circ}{R + j\omega L}$$

The current may be transformed to the time domain by first writing it in polar form:

$$\mathbf{I} = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \angle [-\tan^{-1}(\omega L/R)]$$

and then following the familiar sequence of steps to obtain in a very simple manner the same result we obtained the "hard way" earlier in this chapter.

Example 9.6 KVL in the Frequency Domain (Textbook p. 321)

A $90\ \Omega$ resistor, a $32\ \text{mH}$ inductor, and a $5\ \mu\text{F}$ capacitor are connected in series across the terminals of a sinusoidal voltage source, as shown in Fig. 9.15. The steady-state expression for the source voltage v_s is $750 \cos(5000t + 30^\circ)\ \text{V}$.

- Construct the frequency-domain equivalent circuit.
- Calculate the steady-state current i by the phasor method.

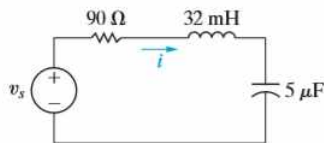


Figure 9.15 ▲ The circuit for Example 9.6.

Solution

- From the expression for v_s , we have $\omega = 5000\ \text{rad/s}$. Therefore the impedance of the $32\ \text{mH}$ inductor is

$$Z_L = j\omega L = j(5000)(32 \times 10^{-3}) = j160\ \Omega,$$

and the impedance of the capacitor is

$$Z_C = j \frac{-1}{\omega C} = -j \frac{10^6}{(5000)(5)} = -j40\ \Omega.$$

The phasor transform of v_s is

$$\mathbf{V}_s = 750 \angle 30^\circ\ \text{V}.$$

Figure 9.16 illustrates the frequency-domain equivalent circuit of the circuit shown in Fig. 9.15.

- We compute the phasor current simply by dividing the voltage of the voltage source by the equivalent impedance between the terminals a,b. From Eq. 9.45,

$$\begin{aligned} Z_{ab} &= 90 + j160 - j40 \\ &= 90 + j120 = 150 \angle 53.13^\circ\ \Omega. \end{aligned}$$

Thus

$$\mathbf{I} = \frac{750 \angle 30^\circ}{150 \angle 53.13^\circ} = 5 \angle -23.13^\circ\ \text{A}.$$

We may now write the steady-state expression for i directly:

$$i = 5 \cos(5000t - 23.13^\circ)\ \text{A}.$$

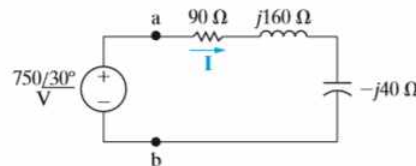
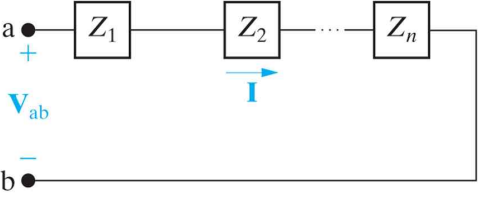
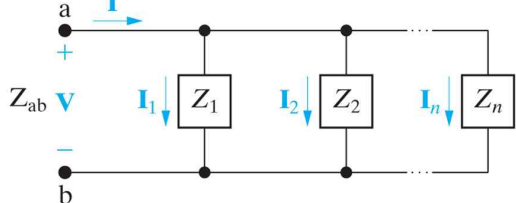


Figure 9.16 ▲ The frequency-domain equivalent circuit of the circuit shown in Fig. 9.15.

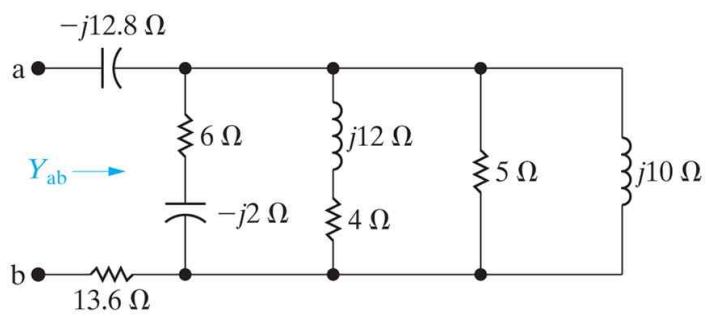
※ How can we get $i(t)$ in the time domain?

§9.6 Series and Parallel Simplifications

series	parallel
	
$V_{ab} = Z_1 I + Z_2 I + \dots + Z_n I$ $= (Z_1 + Z_2 + \dots + Z_n) I$	$I_{ab} = I_1 + I_2 + \dots + I_n$ $\frac{V}{Z_{ab}} = \frac{V}{Z_1} + \frac{V}{Z_2} + \dots + \frac{V}{Z_n}$
$Z_{ab} = \frac{V_{ab}}{I} = Z_1 + Z_2 + \dots + Z_n$	$\frac{1}{Z_{ab}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_n}$

※ Admittance Y [unit: siemens] : $Y = \frac{1}{Z}$

(Example) Z_{ab}



(Example) Combining Impedances in Series and in Parallel

The sinusoidal current source in the circuit shown in Fig. 9.18 produces the current $i_s = 8 \cos 200,000t$ A.

- Construct the frequency-domain equivalent circuit.
- Find the steady-state expressions for v , i_1 , i_2 , and i_3 .

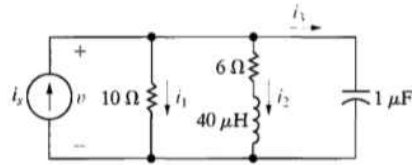


Figure 9.18 ▲ The circuit for Example 9.7.

Solution

- The phasor transform of the current source is $8 \angle 0^\circ$; the resistors transform directly to the frequency domain as 10 and 6 Ω ; the 40 μH inductor has an impedance of $j8 \Omega$ at the given frequency of 200,000 rad/s; and at this frequency the 1 μF capacitor has an impedance of $-j5 \Omega$. Figure 9.19 shows the frequency-domain equivalent circuit and symbols representing the phasor transforms of the unknowns.
- The circuit shown in Fig. 9.19 indicates that we can easily obtain the voltage across the current source once we know the equivalent impedance of the three parallel branches. Moreover, once we know $\mathbf{V}$, we can calculate the three phasor currents $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_3$ by using Eq. 9.35. To find the equivalent impedance of the three branches, we first find the equivalent admittance simply by adding the admittances of each branch. The admittance of the first branch is

$$Y_1 = \frac{1}{10} = 0.1 \text{ S},$$

the admittance of the second branch is

$$Y_2 = \frac{1}{6 + j8} = \frac{6 - j8}{100} = 0.06 - j0.08 \text{ S},$$

and the admittance of the third branch is

$$Y_3 = \frac{1}{-j5} = j0.2 \text{ S}.$$

The admittance of the three branches is

$$\begin{aligned} Y &= Y_1 + Y_2 + Y_3 \\ &= 0.16 + j0.12 \\ &= 0.2 \angle 36.87^\circ \text{ S}. \end{aligned}$$

The impedance at the current source is

$$Z = \frac{1}{Y} = 5 \angle -36.87^\circ \Omega.$$

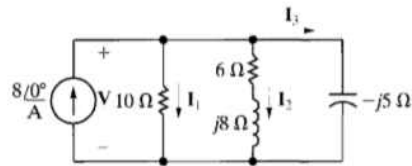


Figure 9.19 ▲ The frequency-domain equivalent circuit.

The Voltage $\mathbf{V}$ is

$$\mathbf{V} = \mathbf{Z}\mathbf{I} = 40 \angle -36.87^\circ \text{ V}.$$

Hence

$$\mathbf{I}_1 = \frac{40 \angle -36.87^\circ}{10} = 4 \angle -36.87^\circ = 3.2 - j2.4 \text{ A},$$

$$\mathbf{I}_2 = \frac{40 \angle -36.87^\circ}{6 + j8} = 4 \angle -90^\circ = -j4 \text{ A},$$

and

$$\mathbf{I}_3 = \frac{40 \angle -36.87^\circ}{5 \angle -90^\circ} = 8 \angle 53.13^\circ = 4.8 + j6.4 \text{ A}.$$

We check the computations at this point by verifying that

$$\mathbf{I}_1 + \mathbf{I}_2 + \mathbf{I}_3 = \mathbf{I}.$$

Specifically,

$$3.2 - j2.4 - j4 + 4.8 + j6.4 = 8 + j0.$$

The corresponding steady-state time-domain expressions are

$$v = 40 \cos(200,000t - 36.87^\circ) \text{ V},$$

$$i_1 = 4 \cos(200,000t - 36.87^\circ) \text{ A},$$

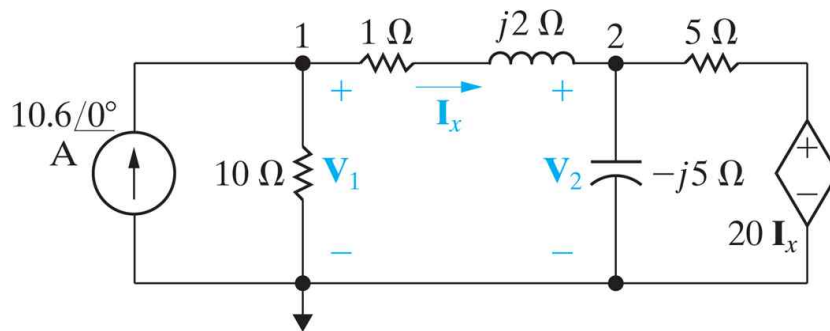
$$i_2 = 4 \cos(200,000t - 90^\circ) \text{ A},$$

$$i_3 = 8 \cos(200,000t + 53.13^\circ) \text{ A}.$$

§9.7 The Node-Voltage Method in Frequency Domain

(Example 9.11) [Textbook p.330]

Using the node-voltage method in the frequency domain



$$-10.6 + \frac{V_1}{10} + \frac{V_1 - V_2}{1 + j2} = 0$$

$$\frac{V_2 - V_1}{1 + j2} + \frac{V_2}{-j5} + \frac{V_2 - 20I_x}{5} = 0$$

$$I_x = \frac{V_1 - V_2}{1 + j2}$$

Solving the above 3 complex number equations, we can get

$$\therefore V_1 = 68.40 - j16.80 \text{ [V]}$$

$$V_2 = 68 - j26 \text{ [V]}$$

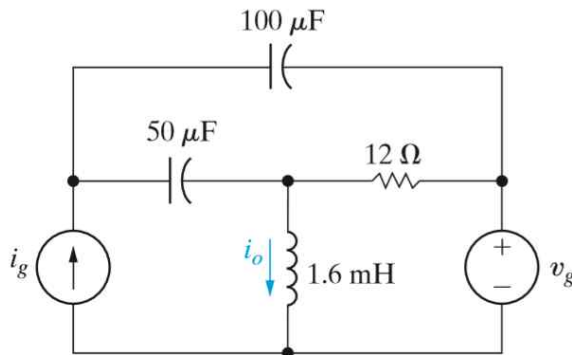
$$I_x = 3.76 + j1.68 \text{ [A]}$$

Problem 9.56 (Textbook p.353)

9.56 Use the node voltage method to find the steady-state expression for i_o in the circuit seen in Fig. P9.56 if $i_g = 5 \cos 2500t$ A and $v_g = 20 \cos(2500t + 90^\circ)$ V.

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Figure P9.56

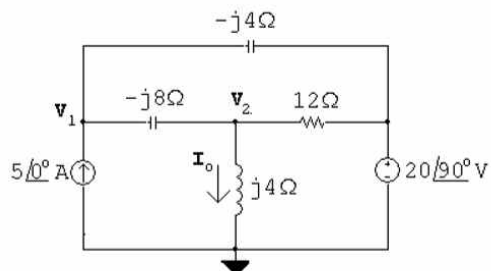


$$P\ 9.56 \quad j\omega L = j(2500)(1.6 \times 10^{-3}) = j4\ \Omega$$

$$\frac{1}{j\omega C} = \frac{-j}{(2500)(100 \times 10^{-6})} = -j4\ \Omega$$

$$\mathbf{I}_g = 5\angle 0^\circ\ \text{A}$$

$$\mathbf{V}_g = 20\angle 90^\circ\ \text{V}$$



$$-5 + \frac{\mathbf{V}_1 - \mathbf{V}_2}{-j8} + \frac{\mathbf{V}_1 - j20}{-j4} = 0$$

$$\frac{\mathbf{V}_2 - \mathbf{V}_1}{-j8} + \frac{\mathbf{V}_2}{j4} + \frac{\mathbf{V}_2 - j20}{12} = 0$$

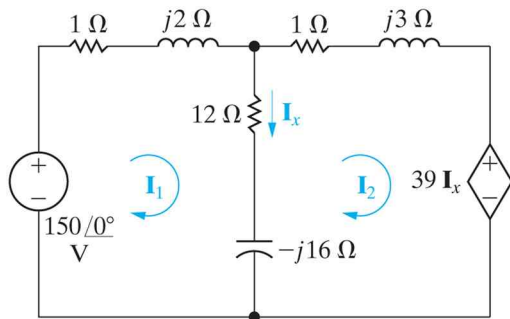
Solving,

$$\mathbf{V}_2 = -8 + j4\ \text{V}; \quad \mathbf{I}_o = \frac{\mathbf{V}_2}{j4} = 1 + j2 = 2.24\angle 63.43^\circ\ \text{A}$$

$$i_o = 2.24 \cos(2500t + 63.43^\circ)\ \text{A}$$

§9.8 The Mesh-Current Method in Frequency Domain

(Example)



$$150 = (1 + j2) I_1 + (12 - j16) (I_1 - I_2)$$

$$0 = (12 - j16) (I_2 - I_1) + (1 + j3) I_2 + 39I_x$$

$$I_x = I_1 - I_2$$

Rearrange

$$150 = (13 - j14) I_1 - (12 - j16) I_2$$

$$0 = -(12 - j16) I_1 + (13 - j13) I_2 + 39I_x$$

$$I_x = I_1 - I_2$$

Finally, we can get

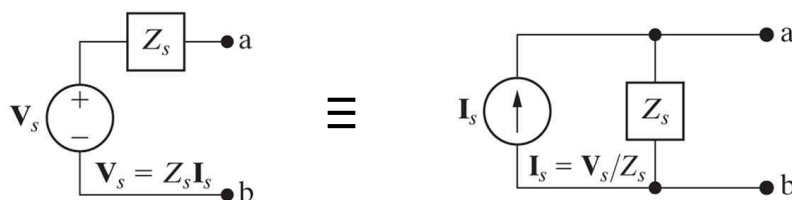
$$I_1 = -26 - j52 \text{ [A]}$$

$$I_2 = -24 - j58 \text{ [A]}$$

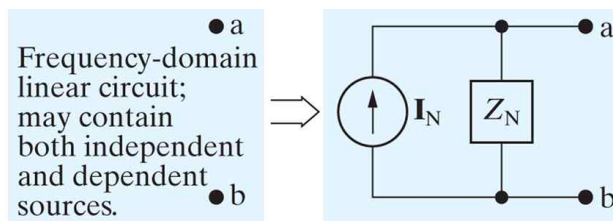
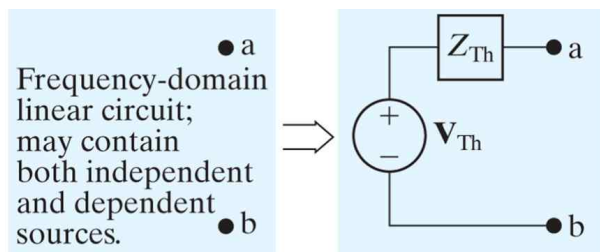
$$I_x = -2 + j6 \text{ [A]}$$

§9.9 Source Transformation, Thevenin Equivalent Circuit

- A source transformation in the frequency domain



- Thevenin/Norton equivalent circuits in the frequency domain



- ※ Circuit analysis techniques in the frequency domain (i.e., AC cases) are same as circuit analysis techniques in the time domain (i.e., DC cases) except complex number calculations.

Summary

- Euler's Theorem
- time domain / frequency domain
- differential equations / complex number algebraic equations
- phasor transformation
- time delay / phase difference
- impedances

	time domain	frequency domain
resistance	$R \ [\Omega]$	$R \ [\Omega]$
inductance	$L \ [\text{H}]$	$j\omega L \ [\Omega]$
capacitance	$C \ [\text{F}]$	$\frac{1}{j\omega C} \ [\Omega]$